

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250287574

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Song; Hoju et al.

SEMICONDUCTOR MEMORY DEVICE AND METHOD OF MANUFACTURING THE SAME

Abstract

A method of manufacturing a semiconductor memory device includes forming bit line structures extending in a first horizontal direction on a substrate, and insulating spacer structures covering opposite sidewalls of each bit line structure, forming a preliminary buried contact material layer and a mold layer to respectively fill lower and upper portions of a space between a pair of insulating spacer structures, patterning the mold layer and the preliminary buried contact material layer into mold patterns spaced apart from each other in a second horizontal direction and buried contacts spaced apart from each other in the second horizontal direction, forming insulating fences among the mold patterns separated from each other and among the buried contacts separated from each other, removing the mold patterns to expose the buried contacts, and forming landing pads on the exposed buried contacts, each landing pad connected to a corresponding one of the exposed buried contacts.

Inventors: Song; Hoju (Seongnam-si, KR), Kim; Seokhyun (Incheon, KR), Kim; Youngjun (Hwaseong-si, KR), Park; Jinhyung (Bucheon-si, KR), Lee; Hyeran (Hwaseong-si, KR), Kim; Bongsoo (Yongin-si, KR), Kim; Sungwoo (Hwaseong-si, KR)

Applicant: Samsung Electronics Co., Ltd. (Suwon-si, KR)

Family ID: 74260541

Assignee: Samsung Electronics Co., Ltd. (Suwon-si, KR)

Appl. No.: 19/219396

Filed: May 27, 2025

Foreign Application Priority Data

KR

10-2019-0092003

Jul. 29, 2019

Related U.S. Application Data

parent US continuation 17512031 20211027 parent-grant-document US 12336171 child US 19219396

parent US continuation 16826655 20200323 parent-grant-document US 11183500 child US 17512031

Publication Classification

Int. Cl.: H10B12/00 (20230101)

U.S. Cl.:

CPC H10B12/485 (20230201); H10B12/0335 (20230201); H10B12/053 (20230201); H10B12/315 (20230201); H10B12/34 (20230201);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. application Ser. No. 17/512,031 filed on Oct. 27, 2021, which is a continuation application of U.S. application Ser. No. 16/826,655 which claims the benefit of Korean Patent Application No. 10-2019-0092003, filed on Jul. 29, 2019, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated herein in its entirety by reference.

BACKGROUND

[0002] The inventive concept relates to a semiconductor memory device and a method of manufacturing the same, and more particularly, to a semiconductor memory device including an insulating fence between contact structures and a method of manufacturing the semiconductor memory device.

[0003] In accordance with the rapid development of the electronics industry and user demand, electronic devices have become more compact and lightweight. Accordingly, a high degree of integration is required for semiconductor memory devices used in electronic devices, and design rules for configurations of semiconductor memory devices are reduced. Highly scaled-down semiconductor memory devices are desirable to ensure electrical reliability of contact structures that connect between active regions and capacitors.

SUMMARY

[0004] The inventive concept provides a semiconductor memory device including contact structures having improved electrical reliability and a manufacturing method of the semiconductor memory device.

[0005] According to an exemplary embodiment of the present invention, a method of manufacturing a semiconductor memory device includes forming a plurality of bit line structures extending in parallel with each other in a first horizontal direction on a substrate, and a plurality of insulating spacer structures covering opposite sidewalls of each of the plurality of bit line structures, forming a preliminary buried contact material layer and a mold layer to respectively fill a lower portion of a space between a pair of insulating spacer structures of the plurality of insulating spacer structures and an upper portion of the space, the pair of insulating spacer structures facing each other across the space, patterning the mold layer and the preliminary buried contact material layer into a plurality of mold patterns spaced apart from each other in a second horizontal direction different from the first horizontal direction and a plurality of buried contacts spaced apart from each other in the second horizontal direction, respectively, forming a plurality of insulating fences among the plurality of mold patterns separated from each other and among the

plurality of buried contacts separated from each other, removing the plurality of mold patterns to expose the plurality of buried contacts, and forming a plurality of landing pads on the plurality of exposed buried contacts, each of the plurality of landing pads connected to a corresponding one of the plurality of exposed buried contacts.

[0006] According to an exemplary embodiment of the present invention, a method of manufacturing a semiconductor memory device includes forming, on a substrate, a plurality of bit line structures, a plurality of first insulating spacer structures each of which covers a first sidewall of a corresponding one of the plurality of bit line structures, and a plurality of second insulating spacer structures each of which covers a second sidewall, opposite the first sidewall, of the corresponding one of the plurality of bit line structures, forming a preliminary buried contact material layer and a mold layer on the substrate, wherein the preliminary buried contact material layer and the mold layer are formed in a first space between one of the plurality of first insulating spacer structures and one of the plurality of second insulating spacer structures adjacent thereto in a first horizontal direction, wherein the preliminary buried contact material layer fills a lower portion of the first space and the mold layer fills an upper portion thereof, and patterning the mold layer and the preliminary buried contact material layer into a plurality of mold patterns spaced apart from each other in a second horizontal direction different from the first horizontal direction and a plurality of buried contacts spaced apart from each other in the second horizontal direction, forming a plurality of insulating fences among the plurality of buried contacts and the plurality of mold patterns, a first insulating fence of the plurality of insulating fences having a lower portion and an upper portion, the lower portion of the first insulating fence being disposed between a pair of buried contacts, spaced apart from each other in the second horizontal direction, among a plurality of buried contacts and having a first width in the second horizontal direction, and the upper portion of the first insulating fence being disposed between a pair of mold patterns, spaced apart from each other in the second horizontal direction, among a plurality of mold patterns and having a second width in the second horizontal direction different from the first width.

[0007] According to an exemplary embodiment of the present invention, a method of manufacturing a semiconductor memory device includes preparing a substrate having a plurality of active regions defined by an element separation layer, forming a plurality of word line trenches crossing the plurality of active regions and extending in parallel with each other in a first horizontal direction, and a plurality of gate dielectric layers and a plurality of word lines to fill the plurality of word line trenches, forming a plurality of bit line structures each comprising a bit line and an insulating capping line covering the bit line on the substrate, the plurality of bit line structures extending in parallel with each other in a second horizontal direction crossing the first horizontal direction, and a plurality of insulating spacer structures covering opposite sidewalls of each of the bit line structures, forming a first preliminary buried contact material layer to fill a first space between a pair of insulating spacer structures among the plurality of insulating spacer structures, the pair of insulating spacer structures facing each other across the first space, removing an upper portion of the first preliminary buried contact material layer to form a second preliminary buried contact material layer and a second space between the pair of insulating spacer structures, forming a mold layer to fill the second space, forming a plurality of mold patterns and a plurality of buried contacts by patterning the mold layer and the second preliminary buried contact material layer respectively, forming, after removing the plurality of mold patterns, a plurality of landing pads on the plurality of buried contacts, and forming a plurality of capacitors on the plurality of landing pads. The plurality of mold patterns are separated from each other and the plurality of buried contacts are separated from each other. Each of the plurality of buried contacts is connected to a corresponding active region of the plurality of active regions. Each of the plurality of landing pads is connected to a corresponding one of the plurality of buried contacts. Each of the plurality of capacitors includes a bottom electrode connected to a corresponding landing pad of the plurality of landing pads.

[0008] According to an exemplary embodiment of the present invention, a semiconductor memory device includes a substrate having a plurality of active regions defined by an element separation layer, a plurality of bit line structures extending in parallel with each other in a first horizontal direction on the substrate and spaced apart from each other in a second horizontal direction different from the first horizontal direction, each of the plurality of bit line structures comprising a bit line an insulating capping line stacked on the bit line, a plurality of first insulating spacer structures covering first sidewalls of the plurality of bit line structures, a plurality of second insulating spacer structures covering second sidewalls of the plurality of bit line structures, a plurality of insulating fences disposed in a first space between one of the plurality of first insulating spacer structures and one of the plurality of second insulating spacer structures adjacent thereto and spaced apart from each other in the first horizontal direction, a plurality of buried contacts disposed in the first space and spaced apart from each other in the first horizontal direction, each of the plurality of buried contacts filling a lower portion of the first space and being connected to a corresponding one of the plurality of active regions, wherein each of the plurality of buried contacts and each of the plurality of insulating fences are alternately arranged in the first horizontal direction, and a plurality of landing pads spaced apart from each other in the first horizontal direction, each of the plurality of landing pads connected to a corresponding one of the plurality of buried contacts, each of the plurality of landing pads filling an upper portion of the first space and extending to an upper portion of a corresponding one of the plurality of bit line structures. Each of the plurality of insulating fences extends from between a corresponding pair of buried contacts among the plurality of buried contacts to between a corresponding pair of landing pads among the plurality of landing pads. A first width of a lower portion of each of the plurality of insulating fences between the corresponding pair of buried contacts is different from a second width of an upper portion of each of the plurality of insulating fences between the corresponding pair of landing pads. The first width and the second width are measured in the first horizontal direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] Embodiments of the inventive concept will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings in which:

[0010] FIG. **1** is a schematic plan layout of main components of a semiconductor memory device, according to example embodiments of the inventive concept;

[0011] FIGS. **2** through **12** are cross-sectional views sequentially illustrating a method of manufacturing a semiconductor memory device, according to embodiments of the inventive concept;

[0012] FIG. **13** is a cross-sectional view of main components of a semiconductor memory device, according to example embodiments of the inventive concept;

[0013] FIGS. **14** and **15** are cross-sectional views sequentially illustrating a method of manufacturing a semiconductor memory device, according to embodiments of the inventive concept;

[0014] FIG. **16** is a cross-sectional view of main components of a semiconductor memory device, according to example embodiments of the inventive concept;

[0015] FIGS. **17** and **18** are cross-sectional views sequentially illustrating a method of manufacturing a semiconductor memory device, according to embodiments of the inventive concept;

[0016] FIG. **19** is a cross-sectional view of main components of a semiconductor memory device, according to example embodiments of the inventive concept;

[0017] FIGS. **20** and **21** are cross-sectional views sequentially illustrating a method of

manufacturing a semiconductor memory device, according to embodiments of the inventive concept;

[0018] FIG. **22** is a cross-sectional view of main components of a semiconductor memory device, according to example embodiments of the inventive concept;

[0019] FIG. **23** is a cross-sectional view of main components of a semiconductor memory device, according to example embodiments of the inventive concept; and

[0020] FIG. **24** is a cross-sectional view of main components of a semiconductor memory device, according to example embodiments of the inventive concept.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0021] FIG. **1** is a schematic plan layout of a semiconductor memory device **10**, according to example embodiments of the inventive concept.

[0022] Referring to FIG. **1**, the semiconductor memory device **10** may include a plurality of active regions ACT. In some embodiment, the plurality of active regions ACT may have a long axis in a diagonal direction with respect to a first horizontal direction (X direction) and a second horizontal direction (Y direction).

[0023] A plurality of word lines WL may extend in parallel with each other in the first horizontal direction (X direction) across the plurality of active regions ACT. On the plurality of word lines WL, a plurality of bit lines BL may extend in parallel with each other in the second horizontal direction (Y direction) across the first horizontal direction (X direction).

[0024] The plurality of bit lines BL may be connected to the plurality of active regions ACT via direct contacts DC.

[0025] In some embodiment, a plurality of buried contacts BC may be formed between two adjacent bit lines BL among the plurality of bit lines BL. In some embodiment, the plurality of buried contacts BC may be arranged in a line in the first horizontal direction (X direction) and the second horizontal direction (Y direction), respectively.

[0026] A plurality of landing pads LP may be formed on the plurality of buried contacts BC. The plurality of landing pads LP may at least partially overlap the plurality of buried contacts BC. In some embodiment, the plurality of landing pads LP may extend to a top portion of any one of the two bit lines BL adjacent to each other. For example, each of the landing pads LP may be disposed between corresponding two bit lines adjacent to each other, overlapping one of the corresponding two bit lines.

[0027] A plurality of storage nodes SN may be formed on the plurality of landing pads LP. The plurality of storage nodes SN may be formed on the plurality of bit lines BL. Each of the plurality of storage nodes SN may be a bottom electrode of each of the plurality of capacitors. The storage node SN may be connected to the active region ACT via the landing pad LP and the buried contact BC.

[0028] Each of FIGS. **2** through **24** illustrates cross-section views taken along lines A-A', B-B', and C-C' in FIG. **1**.

[0029] FIGS. **2** through **12** are cross-sectional views sequentially illustrating a manufacturing method of the semiconductor memory device **10**, according to embodiments of the inventive concept.

[0030] Referring to FIG. **2**, an element isolation trench **116T** may be formed in a substrate **110**, and an element isolation layer **116** may be formed to fill the element isolation trench **116T**. A plurality of active regions **118** may be defined in the substrate **110** by the element isolation layer **116**. The active region **118** may have a relatively long island shape having a short axis and a long axis, like the active region ACT illustrated in FIG. **1**.

[0031] The substrate **110** may include silicon (Si) such as crystalline Si, polycrystalline Si, or amorphous Si. Alternatively, the substrate **110** may include at least one compound semiconductor of a semiconductor element such as germanium (Ge), silicon germanium (SiGe), silicon carbide (SiC), gallium arsenide (GaAs), indium arsenide (InAs), and indium phosphide (InP). Alternatively,

the substrate **110** may have a silicon on insulator (SOI) structure. For example, the substrate **110** may include a buried oxide (BOX) layer. The substrate **110** may include a conductive region, for example, a well doped with impurities, or a structure doped with impurities.

[0032] The element isolation layer **116** may include a material including at least one of, for example, a silicon oxide layer, a silicon nitride layer, and a silicon oxynitride layer. The element isolation layer **116** may include a single layer including one kind of an insulating layer, a double layer including two kinds of insulating layers, or a multiple layer including a combination of at least three kinds of insulating layers. For example, the element isolation layer **116** may include a double layer or a multiple layer of an oxide layer and a nitride layer. However, according to the technical idea of the inventive concept, a configuration of the element isolation layer **116** is not limited thereto.

[0033] A plurality of word line trenches **120T** may be formed in the substrate **110**. The plurality of word line trenches **120T** may extend in parallel with each other in the first horizontal direction (X direction), and may have a line shape in which each of the plurality of word line trenches **120T** is arranged to cross the active region **118** and the plurality of word line trenches **120T** have substantially equal intervals therebetween in the second horizontal direction (Y direction). As illustrated in the cross-sectional view of the line B-B', a step may be formed on a bottom surface of each of the plurality of word line trenches **120T**. In some embodiment, in the plurality of word line trenches **120T**, the element isolation layer **116** and the substrate **110** may be etched by separate etching processes such that an etching depth of the element isolation layer **116** is different from an etching depth of the substrate **110**. In some embodiment, the plurality of word line trenches **120T** may be etched together with the element isolation layer **116** and the substrate **110**, but due to a difference in etching rates between the element isolation layer **116** and the substrate **110**, the etching depth of the element separation layer **116** may be different from the etching depth of the substrate **110** and the step may be formed on the bottom surface of each of the plurality of word line trenches **120T**.

[0034] After cleaning a resultant product in which the plurality of word line trenches **120T** have been formed, a plurality of gate dielectric layers **122**, a plurality of word lines **120**, and a plurality of buried insulating layers **124** may be sequentially formed in the plurality of word line trenches **120T**. The plurality of word lines **120** may constitute the plurality of word lines WL illustrated in FIG. 1.

[0035] The plurality of word lines **120** may fill lower portions inside the plurality of word line trenches **120T**, and the plurality of buried insulating layers **124** may cover the plurality of word lines **120** and fill upper portions inside the plurality of word line trenches **120T**. Thus, the plurality of word lines **120** may extend in parallel with each other in the first horizontal direction (X direction), and may have a line shape in which each of the plurality of word lines **120** is arranged to cross the active region **118** and the plurality of word lines **120** have substantially equal intervals therebetween in the second horizontal direction (Y direction). Likewise, the plurality of buried insulating layer **124** may extend in parallel with each other in the first horizontal direction (X direction), and may have a line shape in which each of the plurality of buried insulating layer **124** is arranged to cross the active region **118** and the plurality of buried insulating layer **124** have substantially equal intervals therebetween in the second horizontal direction (Y direction).

[0036] In some embodiment, the plurality of word lines **120** may include Ti, TiN, Ta, TaN, W, WN, TiSiN, WSiN, or a combination thereof. In some embodiments, each of the plurality of word lines **120** may include a core layer, and a barrier layer between the core layer and the gate dielectric layer **122**. For example, the core layer may include a metal material such as W or a conductive metal nitride such as WN, TiSiN, and WSiN, and the barrier layer may include a metal material such as Ti and Ta or a conductive metal nitride such as TiN and TaN.

[0037] The gate dielectric layer **122** may include at least one of a silicon oxide layer, a silicon nitride layer, a silicon oxynitride layer, an oxide/nitride/oxide (ONO) layer, or a high dielectric

layer having a higher dielectric constant than the silicon oxide layer. For example, the gate dielectric layer **122** may have a dielectric constant of about 10 to about 25. In some embodiments, the gate dielectric layer **122** may include at least one of hafnium oxide (HfO), hafnium silicate (HfSiO), hafnium oxynitride (HfON), hafnium silicon oxynitride (HfSiON), lanthanum oxide (LaO), lanthanum aluminum oxide (LaAlO), zirconium oxide (ZrO), zirconium silicate (ZrSiO), zirconium oxynitride (ZrON), zirconium silicon oxynitride (ZrSiON), tantalum oxide (TaO), titanium oxide (TiO), barium strontium titanium oxide (BaSrTiO), barium titanium oxide (BaTiO), strontium titanium oxide (SrTiO), yttrium oxide (YO), aluminum oxide (AlO), and lead scandium tantalum oxide (PbScTaO). For example, the gate dielectric layer **122** may include HfO.sub.2, Al.sub.2O.sub.3, HfAlO.sub.3, Ta.sub.2O.sub.3, or TiO.sub.2.

[0038] Top surfaces of the plurality of buried insulating layers **124** may be at substantially the same level as top surfaces of the substrate **110**. The buried insulating layer **124** may include one material layer of a silicon oxide layer, a silicon nitride layer, a silicon oxynitride layer, and a combination thereof.

[0039] A top surface of each of the plurality of word lines **120** may be at a level lower than the top surface of the substrate **110**. Bottom surfaces of the plurality of word lines **120** may have a concave-convex shape, and the plurality of active regions **118** may include saddle type fin field effect transistors (saddle type FinFET).

[0040] In the present specification, a level may denote a height with respect to a main surface of the substrate **110** in a vertical direction (Z direction). In other words, that a position is at an identical level or a constant level may denote that a position is at an identical height or a constant height with respect to the main surface of the substrate **110** in the vertical direction (Z direction), and that a position is at a lower/higher level may denote that a position is at a less/greater height with respect to the reference surface (e.g., upper surface) of the substrate **110** in the vertical direction (Z direction).

[0041] In some embodiments, after the plurality of word lines **120** are formed, impurity ions may be implanted into portions of the active regions **118** of the substrate **110** on both sides of the plurality of word lines **120**, and a source region and a drain region may be formed inside the plurality of active regions **118**. In some other embodiment, an impurity ion implantation process for forming the source region and the drain region may be performed before the plurality of word lines **120** are formed.

[0042] Referring to FIG. 3, insulating layer patterns (**112** and **114**), which cover the isolation layer **116**, the plurality of active regions **118**, and the plurality of buried insulating layers **124**, may be formed. For example, the insulating layer patterns (**112** and **114**) may include a silicon oxide layer, a silicon nitride layer, a silicon oxynitride layer, or a combination thereof.

[0043] In some embodiment, the insulating layer patterns (**112** and **114**) may be formed by stacking a plurality of insulating layers including a first insulating layer pattern **112** and a second insulating layer pattern **114**. For example, the second insulating layer pattern **114** may have a greater dielectric constant than that of the first insulating layer pattern **112**.

[0044] In some embodiment, the first insulating layer pattern **112** may include a silicon oxide layer, and the second insulating layer pattern **114** may include a silicon nitride layer.

[0045] In some embodiment, the first insulating layer pattern **112** may include a non-metal dielectric layer, and the second insulating layer pattern **114** may include a metal dielectric layer. For example, the first insulating pattern **112** may include a silicon oxide layer, a silicon nitride layer, a silicon oxynitride layer, or a combination thereof. For example, the second insulating layer **114** may include at least one of hafnium oxide (HfO), hafnium silicate (HfSiO), hafnium oxynitride (HfON), hafnium silicon oxynitride (HfSiON), lanthanum oxide (LaO), lanthanum aluminum oxide (LaAlO), zirconium oxide (ZrO), zirconium silicate (ZrSiO), zirconium oxynitride (ZrON), zirconium silicon oxynitride (ZrSiON), tantalum oxide (TaO), titanium oxide (TiO), barium strontium titanium oxide (BaSrTiO), barium titanium oxide (BaTiO), strontium titanium oxide

(SrTiO), yttrium oxide (YO), aluminum oxide (AlO), and lead scandium tantalum oxide (PbScTaO).

[0046] Next, a direct contact hole **134H** penetrating the first insulating layer pattern **112** and the second insulating layer pattern **114** may be formed. The direct contact hole **134H** may be formed to expose the source region in the active region **118**. In some embodiment, the direct contact hole **134H** may extend into the active region **118**, that is, into the source region.

[0047] Referring to FIG. **4**, a direct contact conductive layer filling the direct contact hole **134H** and covering the first insulating layer pattern **112** and the second insulating layer pattern **114** may be formed. The direct contact conductive layer may include silicon (Si), germanium (Ge), tungsten (W), tungsten nitride (WN), cobalt (Co), nickel (Ni), aluminum (Al), and molybdenum (Mo), Ruthenium (Ru), titanium (Ti), titanium nitride (TiN), tantalum (Ta), tantalum nitride (Ta_N), copper (Cu), or a combination thereof. In some embodiment, the direct contact conductive layer may include an epitaxial silicon layer. In some embodiment, the direct contact conductive layer may include doped polysilicon.

[0048] Next, a metal-based conductive layer which covers the first insulating layer pattern **112**, the second insulating layer pattern **114**, and the direct contact conductive layer, forms a bit line structure **140**, and an insulating capping layer may be sequentially formed.

[0049] In some embodiment, the metal-based conductive layer may have a stacked structure including a first metal-based conductive layer and a second metal-based conductive layer. The metal-based conductive layer may have a conductive layer stacked structure of a double layer structure, but this is an example, and the technical idea of the inventive concept is not limited thereto. For example, the metal-based layer may have a plurality of stacked structures of a single layer or three or more layers.

[0050] In some embodiment, the first metal-based conductive layer may include titanium nitride (TiN) or Ti—Si—N (TSN), and the second metal-based conductive layer may include W, or tungsten silicide (WSi_x). In some embodiment, the first metal-based conductive layer may function as a diffusion barrier. In some embodiment, the insulating capping layer may include a silicon nitride layer.

[0051] By etching the first metal-based conductive layer, the second metal-based conductive layer, and the insulating capping layer, a plurality of bit lines **147** including the first metal-based conductive pattern **145** and the second metal-based conductive pattern **146** in a line shape, and a plurality of insulating capping lines **148** may be formed. One bit line **147** and one insulating capping line **148** covering the one bit line **147** may constitute one bit line structure **140**.

[0052] In some embodiment, the bit line structure **140** may further include a conductive semiconductor pattern **132** between the second insulating layer pattern **114** and the first metal-based conductive pattern **145**. The conductive semiconductor pattern **132** may include doped polysilicon, for example. In some embodiment, the conductive semiconductor pattern **132** may not be formed but omitted.

[0053] Each of the plurality of bit line structures **140** including the plurality of bit lines **147** and the plurality of insulating capping lines **148** may extend in parallel with each other and in parallel with the main surface of the substrate **110** in the second horizontal direction (Y direction). The plurality of bit lines **147** may constitute the plurality of bit lines BL illustrated in FIG. **1**.

[0054] In an etching process of forming the plurality of bit lines **147**, the plurality of direct contact conductive pattern **134** may be formed by removing together with a portion of the direct contact conductive layer that does not vertically overlap the bit line **147**. In this case, the first insulating layer pattern **112** and the second insulating layer pattern **114** may function as etch stop layers in an etching process of forming the plurality of bit lines **147** and the plurality of direct contact conductive patterns **134**. The plurality of direct contact conductive patterns **134** may constitute the plurality of direct contacts DC illustrated in FIG. **1**. The plurality of bit lines **147** may be electrically connected to the plurality of active regions **118** via the plurality of direct contact

conductive patterns **134**.

[0055] In some embodiment, the conductive semiconductor pattern **132** and the direct contact conductive pattern **134** may be formed together from the direct contact conductive layer in an etching process. For example, the conductive semiconductor pattern **132** may be a portion that does not vertically overlap the direct contact hole **134H** and is on the first insulating layer pattern **112** and the second insulating layer pattern **114** among the portions of the direct contact conductive layer vertically overlapping the bit line **147**, and the direct contact conductive pattern **134** may be a portion that vertically overlaps the direct contact hole **134H** and contact the active region **118**. The term “contact,” as used herein, refers to a direction connection (i.e., touching) unless the context indicates otherwise.

[0056] Both sidewalls of each of the plurality of bit line structures **140** may be covered by an insulating spacer structure **150**. Each of the plurality of insulating spacer structures **150** may include a first insulating spacer **152**, a second insulating spacer **154**, and a third insulating spacer **156**. The second insulating spacer **154** may include a material having a lower dielectric constant than that of the first insulating spacer **152** and that of the third insulating spacer **156**. In some embodiment, the first insulating spacer **152** and the third insulating spacer **156** may include a nitride layer, and the second insulating spacer **154** may include an oxide layer. In some embodiment, the first insulating spacer **152** and the third insulating spacer **156** may include a nitride layer, and the second insulating spacer **154** may include a material having etching selectivity with respect to that of the first insulating spacer **152** and that of the third insulating spacer **156**. For example, when the first insulating spacer **152** and the third insulating spacer **156** include a nitride layer, the second insulating spacer **154** may include an oxide layer, but may be removed in a subsequent process and be replaced with an air spacer. The term “air” as discussed herein, may refer to atmospheric air, or other gases that may be present during the manufacturing process.

[0057] A plurality of buried contact holes **170H** may be formed between the plurality of bit lines **147**. The plurality of buried contact holes **170H** may have an inside space limited by the insulating spacer structure **150** that covers sidewalls of each of two neighboring bit lines **147** between two neighboring bit lines **147** among the plurality of bit lines **147**, and by the active region **118**.

[0058] The plurality of buried contact holes **170H** may be formed by removing portions of the first insulating layer pattern **112** and the second insulating layer pattern **114** by using the plurality of insulating capping lines **148** and insulating spacer structures **150** that covers both sidewalls of each of the plurality of bit line structures **140** as etching masks. After performing first an anisotropic etching process of removing portions of the first insulating layer pattern **112** and the second insulating layer pattern **114** by using the plurality of insulating capping lines **148** and insulating spacer structures **150** that covers both sidewalls of each of the plurality of bit line structures **140** as etching masks, and performing an isotropic etching process of further removing other portions of the active region **118**, the plurality of buried contact holes **170H** may be formed to have an expanded space that has been defined by the active region **118**.

[0059] Referring to FIG. 5, a preliminary buried contact material layer **170p** filling the inside of the plurality of buried contact holes **170H** may be formed. For example, the preliminary buried contact material layer **170p** may include polysilicon. After a conductive material layer filling the plurality of buried contact holes **170H** and covering the plurality of bit line structures **140** is formed, the preliminary buried contact material layer **170p** may be formed by performing a planarization process of removing a portion of the conductive material layer to expose the plurality of insulating capping lines **148**. In some embodiment, a planarization process of removing a portion of the conductive material layer may be omitted, and in this case, a level of an uppermost end of the preliminary buried contact material layer **170p** may be higher than a level of an uppermost end of the plurality of insulating capping lines **148**.

[0060] Referring to FIG. 6, by removing an upper portion of the preliminary buried contact material layer **170p** such that a level of the top surface of the preliminary buried contact material

layer **170p** is lower than a level of the uppermost end of the plurality of insulating capping lines **148**, the preliminary buried contact material layer **170p** may fill a lower portion of the buried contact hole **170H**. In some embodiment, the upper portion of the preliminary buried contact material layer **170p** may be partially removed such that the level of the top surface of the preliminary buried contact material layer **170p** is not lower than the level of the top surface of the bit line **147**. However, the embodiment is not limited thereto.

[0061] In some embodiment, processes described with reference to FIGS. **5** and **6** may not be sequentially performed, but may be performed in-situ. For example, after forming the conductive material layer filling the plurality of buried contact holes **170H** and covering the plurality of bit line structures **140**, and removing an upper portion of the conductive material layer, the preliminary buried contact material layer **170p** that fills a lower portion of the space between the plurality of insulating spacer structures **150** covering the sidewalls of each of the plurality of bit line structures **140** may be formed.

[0062] Referring to FIG. **7**, a mold layer **175** that fills the removed portion of the preliminary buried contact material layer **170p** described with reference to FIG. **6** may be formed. A lower portion of each of the plurality of buried contact holes **170H** may be filled by the preliminary buried contact material layer **170p**, and an upper portion of each of the plurality of buried contact holes **170H** may be filled by the mold layer **175**. In other words, a lower portion of the space between the plurality of insulating spacer structures **150** covering both sidewalls of each of the plurality of bit line structures **140** may be filled by the preliminary buried contact material layer **170p**, and the upper portion of the space may be filled by the mold layer **175**.

[0063] In some embodiment, the level of the top surface of the mold layer **175** may be substantially the same as the level of the uppermost end of the plurality of insulating capping lines **148**. For example, the top surface of the mold layer **175** and the top surfaces of the plurality of insulating capping lines **148** may be coplanar.

[0064] The mold layer **175** may include a material having etch selectivity with respect to the insulating capping line **148**, the insulating spacer structure **150**, and the preliminary buried contact material layer **170p**. For example, when the insulating capping line **148** and the insulating spacer structure **150** include a nitride layer, and the preliminary buried contact material layer **170p** includes polysilicon, the mold layer **175** may include an oxide layer, but the embodiment is not limited thereto.

[0065] Referring to FIG. **8**, a plurality of mask patterns MK may be formed on the mold layer **175**. The plurality of mask patterns MK may extend in parallel with each other in the first horizontal direction (X direction). The plurality of mask patterns MK may be at locations where the plurality of mask patterns MK overlap portions between the plurality of word lines **120** in the vertical direction (Z direction). In other words, a space between a pair of adjacent mask patterns MK among the plurality of mask patterns MK may overlap one of the plurality of word lines **120** in the vertical direction (Z direction).

[0066] Referring to FIGS. **8** and **9**, by removing a portion of the mold layer **175** and a portion of the preliminary buried contact material layer **170p** by using the plurality of mask patterns MK as etching masks, a plurality of fence holes **180H** causing the buried insulating layer **124** to be exposed to a bottom surface of each of the plurality of fence holes **180H** may be formed. In some embodiment, the plurality of fence holes **180H** may be arranged in a line in the first horizontal direction (X direction) and the second horizontal direction (Y direction), respectively.

[0067] The preliminary buried contact material layer **170p** may be separated into a plurality of buried contacts **170** by the plurality of fence holes **180H**. In some embodiment, the plurality of buried contacts **170** may be arranged in a line in the first horizontal direction (X direction) and the second horizontal direction (Y direction), respectively. Each of the plurality of buried contacts **170** may extend from the active region **118** in the vertical direction (Z direction in FIG. **16**) perpendicular to the substrate **110**. The plurality of buried contacts **170** may constitute the plurality

of buried contacts BC illustrated in FIG. 1.

[0068] Similarly, the mold layer **175** may be separated into a plurality of patterns by the plurality of fence holes **180H** when the plurality of buried contacts **170** are formed. In some embodiment, the plurality of mold layers **175** may be arranged in a line in the first horizontal direction (X direction) and the second horizontal direction (Y direction), respectively. Each of the plurality of mold layers **175** may extend from the buried contact **170** in the vertical direction (Z direction in FIG. 16) perpendicular to the substrate **110**.

[0069] In the process of forming the plurality of fence holes **180H**, the plurality of insulating capping lines **148** and the plurality of insulating spacer structures **150** together with the plurality of mask patterns MK may function as etching masks.

[0070] Although not separately illustrated, in some embodiment, in the process of forming the plurality of fence holes **180H**, upper portions of the plurality of insulating capping lines **148** and/or upper portions of the plurality of insulating spacer structures **150** may be removed. In some other embodiment, in the forming process of the plurality of fence holes **180H**, when the upper portions of the plurality of insulating capping lines **148** and/or upper portions of the plurality of insulating spacer structures **150** are removed, a level of an uppermost end of the second insulating spacers **154** may be lower than levels of uppermost ends of the first insulating spacer **152** and the third insulating spacer **156**.

[0071] In addition, although not separately illustrated or described, the upper portions of the plurality of insulating capping lines **148** and/or the upper portions of the plurality of insulating spacer structures **150** may be further removed in a subsequent process, but a level of a top end of the plurality of insulating capping lines **148** and a level of a top end of the insulating spacer structure **150** may be maintained higher than a level of a top end of the plurality of bit lines **147**.

[0072] In some embodiment, the plurality of fence holes **180H** may be formed by sequentially performing a first etching process in which the preliminary buried contact material layer **170p** is exposed by removing a portion of the mold layer **175** by using the plurality of mask patterns MK as etching masks, and performing a second etching process in which the exposed portion of the preliminary buried contact material layer **170p** is removed.

[0073] In some embodiment, the plurality of fence holes **180H** may be formed by an etching process in which a portion of the mold layer **175** and a portion of the preliminary buried contact material layer **170p** are removed in-situ by using the plurality of mask patterns MK as etching masks.

[0074] Referring to FIG. 10, a plurality of insulating fences **180** filling the plurality of fence holes **180H** may be formed. The plurality of insulating fences **180** may include, for example, a nitride layer. After a fence material layer filling the plurality of fence holes **180H** and covering a top surface of the mold layer **175** is formed, the plurality of insulating fences **180** may be formed by removing a portion of the fence material layer covering the top surface of the mold layer **175**.

[0075] The plurality of buried contacts **170** may be in spaces defined by the plurality of insulating spacer structures **150** that cover sidewalls of the plurality of bit line structures **140**, and by the plurality of insulating fences **180**. In other words, the plurality of buried contacts **170** and the plurality of insulating fences **180** may be alternately arranged, in the second horizontal direction (Y direction), in the space between a pair of insulating spacer structures **150** that face each other among the plurality of insulating spacer structures **150** covering sidewalls of the plurality of bit line structures **140**. For example, each of the plurality of buried contacts **170** may be disposed in a space defined by two adjacent insulating spacer structures facing each other across the space (e.g., cross-section A-A' of FIG. 10) and by two adjacent insulating fences (e.g., cross-section C-C' of FIG. 10). For example, the plurality of buried contacts **170** may be spaced apart from each other in the first horizontal direction (X direction) by the plurality of insulating spacer structures **150** and spaced apart from each other in the second horizontal direction (Y direction) by the plurality of insulating fences **180**.

[0076] Referring to FIGS. **10** and **11** together, a plurality of landing pad holes **190H** may be formed by removing the mold layer **175**. The plurality of buried contacts **170** may be exposed to bottom surfaces of the plurality of landing pad holes **190H**.

[0077] Referring to FIG. **12**, a plurality of landing pads **190** that fill the plurality of landing pad holes **190H** and extend over the plurality of bit lines **147** may be formed. The plurality of landing pads **190** may be on the plurality of buried contacts **170**, and may extend over the plurality of bit lines **147**. The plurality of landing pads **190** may be on the plurality of buried contacts **170**, and each of the plurality of buried contacts **170** may be electrically connected to a corresponding one of the plurality of landing pads **190**. The plurality of landing pads **190** may be connected to the active region **118** via the plurality of buried contacts **170**. The plurality of landing pads **190** may constitute the plurality of landing pads LP illustrated in FIG. **1**.

[0078] One landing pad **190** on one buried contact **170** together may be referred to as a contact structure. The buried contacts **170** constituting the contact structure may be between two bit line structures **140** adjacent to each other, and the landing pad **190** may be disposed on the buried contact **170** and extend, in a space between two adjacent bit line structures **140**, from the buried contact **170** to one bit line structure **140**. In other words, the landing pad **190** may be electrically connected to the buried contact **170**, and may be formed to extend, in a space between two adjacent bit line structures **140**, from the buried contact **170** to a top portion of one bit line structure **140** such that the landing pad **190** vertically overlaps one bit line structure **140**.

[0079] In an example embodiment, a landing pad material layer may be formed to fill the plurality of landing pad holes **190H**, covering the plurality of insulating capping lines **148** and the plurality of insulating fences **180**. A portion of an upper surface of the landing pad material layer may be recessed to form a recess portion **190R**, thereby forming the plurality of landing pads **190**. The recess portion **190R** may separate the landing pad material layers into the plurality of landing pads **190** each of which is connected to one of the plurality of buried contacts **170**. The plurality of landing pads **190** may be spaced apart from each other with the recess portion **190R** therebetween. In the recess portion **190R**, the top end of the insulating spacer structure **150**, the top end of the insulating capping line **148**, and a top end of the insulating fence **180** may be exposed. In some embodiment, in the forming process of the recess portion **190R**, the upper portion of the plurality of insulating spacer structures **150**, the upper portion of the plurality of insulating capping lines **148**, and an upper portion of the plurality of insulating fences **180** may be removed.

[0080] In some embodiment, a metal silicide layer may be formed on the plurality of buried contacts **170** before the plurality of landing pads **190** is formed. The metal silicide layer may be between the plurality of buried contacts **170** and the plurality of landing pads **190**. The metal silicide layer may include cobalt silicide (CoSi), nickel silicide (NiSi), or manganese silicide (MnSi), but is not limited thereto.

[0081] In some embodiment, the plurality of landing pads **190** may include a conductive barrier layer and a conductive pad material layer on the conductive barrier layer. For example, the conductive barrier layer may include metal, conductive metal nitride, or a combination thereof. In some embodiment, the conductive barrier layer may have a Ti/TiN stacked structure. For example, the conductive pad material layer may include metal. In some embodiment, the conductive pad material layer may include W.

[0082] FIG. **13** is a cross-sectional view of the semiconductor memory device **1**, according to example embodiments of the inventive concept. The semiconductor memory device **1** may correspond to the semiconductor device **10** of FIG. **1**.

[0083] Referring to FIG. **13**, a plurality of capacitor structures **200** may be formed by sequentially forming a plurality of bottom electrodes **210**, a capacitor dielectric layer **220**, and a top electrode **230** on the plurality of landing pads **190**. Each of the plurality of bottom electrodes **210** may be electrically connected to a corresponding one of the plurality of land pads **190**. The capacitor dielectric layer **220** may conformally cover the plurality of bottom electrodes **210**. The top

electrode **230** may cover the capacitor dielectric layer **220**. The top electrode **230** may face the bottom electrode **210** with the capacitor dielectric layer **220** therebetween. Each of the capacitor dielectric layer **220** and the top electrode **230** may be integrally formed to cover the plurality of bottom electrodes **210** together, in a certain region, for example, in one cell block. The plurality of bottom electrodes **210** may constitute the plurality of storage nodes SN illustrated in FIG. 1.

[0084] Each of the plurality of bottom electrodes **210** may have a cylindrical shape with a closed one end, but is not limited thereto. In some embodiment, each of the plurality of bottom electrodes **210** may have a pillar shape, that is, a pillar shape the inside of which is filled to have a circular horizontal cross section. In some embodiment, the plurality of bottom electrodes **210** may be in a honeycomb zigzag shape with respect to the first horizontal direction (X direction) or the second horizontal direction (Y direction). In some other embodiment, the plurality of bottom electrodes **210** may be in a line matrix in the first horizontal direction (X direction) and the second horizontal direction (Y direction), respectively. The plurality of bottom electrodes **210** may include, for example, silicon doped with impurities, metal such as tungsten and copper, or a conductive metal compound such as titanium nitride. Although not separately illustrated, the semiconductor memory device **1** may further include at least one support pattern in contact with sidewalls of the plurality of bottom electrodes **210**.

[0085] The capacitor dielectric layer **220** may include, for example, TaO, TaAlO, TaON, AlO, AlSiO, HfO, HfSiO, ZrO, ZrSiO, TiO, TiAlO, (Ba,Sr)TiO (BST), SrTiO (STO), BaTiO (BTO), Pb(Zr,Ti)O (PZT), (Pb,La)(Zr,Ti)O, Ba(Zr,Ti)O, Sr(Zr,Ti)O, or a combination thereof.

[0086] The top electrode **230** may include, for example, doped silicon, Ru, RuO, Pt, PtO, Ir, IrO, SrRuO (SRO), (Ba,Sr)RuO (BSRO), CaRuO (CRO), BaRuO, La(Sr,Co)O, Ti, TiN, W, WN, Ta, TaN, TiAlN, TiSiN, TaAlN, TaSiN, or a combination thereof.

[0087] Before the plurality of capacitor structures **200** is formed, an insulating structure **195** filling the recess portion **190R** may be formed. In some embodiment, the insulating structure **195** may include an interlayer insulating layer and an etch stop layer. For example, the interlayer insulating layer may include an oxide layer, and the etch stop layer may include a nitride layer. FIG. 13 illustrates that a top surface of the insulating structure **195** and a bottom surface of the bottom electrode **210** are positioned at the same level, but the embodiment is not limited thereto. For example, a level of the top surface of the insulating structure **195** may be higher than a level of the bottom surface of the bottom electrode **210**, and the bottom electrode **210** may extend toward the substrate **110** into the insulating structure **195**.

[0088] The semiconductor memory device **1** may include the substrate **110** having the plurality of active regions **118**. The plurality of gate dielectric layers **122**, the plurality of word lines **120**, and the plurality of buried insulating layers **124** may be sequentially formed inside the plurality of word line trenches **120T** formed in the substrate **110**. The element isolation layer **116** may define the plurality of active regions **118**. The plurality of insulating layer patterns (**112** and **114**) may be formed to cover the plurality of buried insulating layers **124**. The plurality of landing pads **190** may be formed to fill upper portions of first spaces defined by the plurality of insulating spacer structures **150** covering sidewalls of the plurality of bit line structures **140** and upper portions of second spaces between the plurality of insulating fences **180**. The plurality of buried contacts **170** may fill lower portions of the first spaces and lower portions of the second spaces. Each of the plurality of buried contacts may be connected to a corresponding one of the plurality of active regions **118**. Each of the plurality of landing pads may extend to the upper portion of a corresponding one of the plurality of bit line structures **140**. The plurality of capacitor structures **200** may be formed to include the plurality of bottom electrodes **210**, the capacitor dielectric layer **220**, and the top electrode **230**. Each of the bottom electrodes **210** may be connected to a corresponding one of the plurality of landing pads **190**.

[0089] The plurality of insulating fences **180** may be disposed in the space between a pair of insulating spacer structures **150** facing each other among the plurality of insulating spacer

structures **150** covering both sidewalls of the plurality of bit line structures **140** and may be spaced apart from each other in the second horizontal direction (Y direction). Each of the plurality of insulating fences **180** may extend in the vertical direction (Z direction) from a space between the plurality of buried contacts **170** and to a space between the plurality of landing pads **190**.

[0090] When the plurality of insulating fences **180** are first formed by patterning using an etching process before the preliminary buried contact material layer (**170p** in FIG. 5) is formed, the plurality of buried contact holes **170H** may be difficult to form, and thus a distribution of electrical connection characteristics between the plurality of buried contacts **170** and the plurality of active regions **118** may increase in a semiconductor memory device. In addition, since a big loss may occur in the upper portions of the plurality of insulating capping lines **148** and the plurality of insulating spacer structures **150**, a volume of the second insulating spacer **154** including a material of a relatively low permittivity or an air space formed by removing the second insulating spacer **154** may be reduced, and thus a parasitic capacitance between the plurality of bit lines **147** may be increased.

[0091] However, since in the semiconductor memory device **1** according to various embodiments, the plurality of insulating fences **180** are formed after the plurality of buried contacts **170** are formed, electrical connection reliability between the plurality of buried contacts **170** and the plurality of active regions **118** may increase, and thus the distribution of electrical characteristics may decrease in the semiconductor memory device **10**. In addition, since the plurality of insulating fences **180** are formed not by an etching process but by a damascene method of filling the plurality of fence holes **180H**, an occurrence of the loss in the upper portions of the plurality of insulating capping lines **148** and in the upper portions of the plurality of insulating spacer structures **150** may be reduced, and thus an increase in the parasitic capacitance between the plurality of bit lines **147** may be prevented.

[0092] When the plurality of insulating fences **180** are first formed and then the upper portion of the preliminary buried contact material layer **170p** is removed to form the plurality of buried contacts **170**, in the process of forming the space (i.e., the plurality of fence holes **180H**) for the plurality of insulating fences **180** by performing an etching process on the preliminary buried contact material layer **170p**, defects in an etching profile of the space for the plurality of insulating fences **180** may occur.

[0093] However, in the semiconductor memory device **10** according to the inventive concept, after the upper portion of the preliminary buried contact material layer **170p** is removed, and the mold layer **175** filling the removed portion of the upper portion of the preliminary buried contact material layer **170p** is formed, the space for the plurality of insulating fences **180** may be formed by performing an etching process on the mold layer **175** and the preliminary buried contact material layer **170p**. For example, when the mold layer **175** includes an oxide layer and the preliminary buried contact material layer **170p** includes polysilicon, since an etching process is first performed on the mold layer **175** including an oxide layer, the etching profiles of the space (i.e., the plurality of fence holes **180H**) for the insulating fence **180** may provide a better profile for its subsequent process of filling a conductive material in the space, and thus an electrical connection reliability between the plurality of capacitor structures **200** and the plurality of active regions **118** via the plurality of buried contacts **170** and the plurality of landing pads **190** between the plurality of insulating fences **180** may increase.

[0094] FIGS. **14** through **15** are cross-sectional views sequentially illustrating a manufacturing method of the semiconductor memory device **1**, according to embodiments of the inventive concept. FIG. **14** is a cross-sectional view illustrating operations subsequent to an operation with reference to FIG. **6**, and descriptions given with reference to FIGS. **14** and **15** that are duplicates of descriptions given with reference to FIGS. **2** through **12** are omitted, and the same reference numerals may denote the same components.

[0095] Referring to FIGS. **6** and **14** together, the upper portions of the plurality of insulating spacer

structures **150** that are above the upper portions of the preliminary buried contact material layer **170p** may be removed such that the upper portions of the plurality of insulating capping lines **148** protrude from the top surfaces of the plurality of preliminary buried contact material layer **170p**. [0096] Referring to FIG. **15**, a plurality of expansion capping layers **158** that cover surfaces of the plurality of insulating capping lines **148** protruding toward top surfaces of the plurality of preliminary buried contact material layer **170p** may be formed. One bit line **147**, one insulating capping line **148** covering the one bit line **147**, and one expansion capping layer **158** covering the surface of the insulating capping line **148** may constitute one bit line structure **140a**. In an example embodiment, the plurality of expansion capping layers **158** may include or be formed of the same material, such as nitride, as that of the plurality of insulating capping lines **148**.

[0097] The expansion capping layer **158** may cover a top surface of the second insulating spacer **154** on the top end of the plurality of insulating spacer structures **150**. In some embodiment, the second insulating spacer **154** is removed and then the expansion capping layer **158** is formed to cover a top surface of the first insulating spacer **152** and a top surface of the third insulating spacer **156**, thereby forming an air space defined by the first insulating spacer **152**, the third insulating spacer **156**, and the expansion capping layer **158**. The second insulating spacer **154** may be replaced with the air spacer.

[0098] In the expansion capping layer **158**, a width of a portion covering a top surface of the insulation capping line **148** may be greater than a width of a portion covering a side surface of the insulation capping line **148** protruding to the top surface of the preliminary buried contact material layer **170p**. A top end of the bit line structure **140a** may have a relatively large area by the expansion capping layer **158**.

[0099] FIG. **16** is a cross-sectional view of main components of a semiconductor memory device **2**, according to example embodiments of the inventive concept. The semiconductor memory device **2** may correspond to the semiconductor device **10** of FIG. **1**.

[0100] Referring to FIG. **16**, the semiconductor memory device **2** may be formed by performing the processes described with reference to FIGS. **7** through **13**.

[0101] The semiconductor memory device **1** may include the substrate **110** having the plurality of active regions **118**. The plurality of gate dielectric layers **122**, the plurality of word lines **120**, and the plurality of buried insulating layers **124** may be sequentially formed inside the plurality of word line trenches **120T** formed in the substrate **110**. The element isolation layer **116** may be formed to define the plurality of active regions **118**. The plurality of insulating layer patterns (**112** and **114**) may be formed to cover the plurality of buried insulating layers **124**. The plurality of bit line structures **140a** may be formed on the plurality of insulating patterns (**112** and **114**). The plurality of landing pads **190** may be formed to fill first spaces defined by the plurality of insulating spacer structures **150** covering sidewalls of the plurality of bit line structures **140a** and upper portions of second spaces between the plurality of insulating fences **180**. The plurality of buried contacts **170** may fill lower portions of the first spaces and lower portions of the second spaces. Each of the plurality of buried contacts **170** may be connected to a corresponding one of the plurality of active regions **118**. Each of the plurality of landing pads **190** may extend to the upper portion of a corresponding one of the plurality of bit line structures **140**. The plurality of capacitor structures **200** may include the plurality of bottom electrodes **210**, the capacitor dielectric layer **220**, and the top electrode **230**. Each of the plurality of bottom electrodes **210** may be connected to a corresponding one of the plurality of landing pads **190**. The plurality of landing pads **190** may cover at least a portion of a top surface of the plurality of expansion capping layers **158** of the bit line structure **140a**.

[0102] FIGS. **17** and **18** are cross-sectional views sequentially illustrating a manufacturing method of the semiconductor memory device **2**, according to embodiments of the inventive concept. FIG. **17** is a cross-sectional view illustrating operations subsequent to an operation with reference to FIG. **8**, and descriptions given with reference to FIGS. **17** and **18** that are duplicates of descriptions

given with reference to FIGS. 2 through 12 are omitted, and the same reference numerals may denote the same components.

[0103] Referring to FIGS. 8 and 17 together, by removing a portion of the mold layer 175 and a portion of the preliminary buried contact material layer 170p by using the plurality of mask patterns MK as etching masks, a plurality of fence holes 180Ha that causes the buried insulating layer 124 to be exposed to a bottom surface of each of the plurality of fence holes 180Ha may be formed. In some embodiment, the plurality of fence holes 180Ha may be in a line in the first horizontal direction (X direction) and the second horizontal direction (Y direction), respectively.

[0104] The preliminary buried contact material layer 170p may be separated into a plurality of buried contacts 170 by the plurality of fence holes 180Ha.

[0105] In the process of forming the plurality of fence holes 180Ha, due to differences in etching characteristics of the mold layer 175 and the preliminary buried contact material layer 170p, a second width W2 that is a width of an upper portion of the plurality of fence holes 180Ha between a pair of the mold layers in the second horizontal direction (Y direction) may be greater than a first width W1 that is a width of a lower portion of the plurality of fence holes 180Ha between a pair of the buried contacts 170 in the second horizontal direction (Y direction) among the plurality of fence holes 180Ha. In other words, in the plurality of fence holes 180Ha, the first width W1 that is the width of the lower portion in the second horizontal direction (Y direction) may be less than the second width W2 that is the width of the upper portion in the second horizontal direction (Y direction). In some embodiment, the first width W1 may less than the second width W2 by about 1 nm to about 5 nm.

[0106] A width of the plurality of fence holes 180Ha in the first horizontal direction (X direction) may be defined by a width between a pair of insulating spacer structures 150 facing each other, and may increase in the vertical direction (Z direction) perpendicular to the substrate 110.

[0107] In some embodiment, after the plurality of fence holes 180H illustrated in FIG. 9 is formed, a portion of the mold layer 175 may be further removed to form the plurality of fence holes 180Ha illustrated in FIG. 17.

[0108] Referring to FIG. 18, a plurality of insulating fences 180a filling the plurality of fence holes 180Ha may be formed. The plurality of insulating fences 180a may include, for example, a nitride layer. After a fence material layer filling the plurality of fence holes 180Ha and covering a top surface of the mold layer 175 is formed, the plurality of insulating fences 180a may be formed by removing a portion of the fence material layer covering the top surface of the mold layer 175.

[0109] The plurality of buried contacts 170 may be in spaces defined by the plurality of insulating spacer structures 150 that cover sidewalls of the plurality of bit line structures 140, and by the plurality of insulating fences 180a. For example, each of the plurality of buried contacts 170 may be disposed in a space defined by two adjacent insulating spacer structures facing each other across the space (e.g., cross-section A-A' of FIG. 18) and by two adjacent insulating fences (e.g., cross-section C-C' of FIG. 18). For example, the plurality of buried contacts 170 may be spaced apart from each other in the first horizontal direction (X direction) by the plurality of insulating spacer structures 150 and spaced apart from each other in the second horizontal direction (Y direction) by the plurality of insulating fences 180a.

[0110] FIG. 19 is a cross-sectional view of a semiconductor memory device 3, according to example embodiments of the inventive concept. The semiconductor memory device 3 may correspond to the semiconductor device 10 of FIG. 1.

[0111] Referring to FIG. 19, the semiconductor memory device 3 may be formed by performing the processes described with reference to FIGS. 10 through 13 on the resulting structure of FIG. 18.

[0112] The semiconductor memory device 3 may include the substrate 110 having the plurality of active regions 118. The plurality of gate dielectric layers 122, the plurality of word lines 120, and the plurality of buried insulating layers 124 may be sequentially formed inside the plurality of word line trenches 120T formed in the substrate 110. The element isolation layer 116 may be formed to

define the plurality of active regions **118**. The plurality of insulating layer patterns (**112** and **114**) may be formed to cover the plurality of buried insulating layers **124**. The plurality of landing pads **190** may be formed to fill upper portions of first spaces defined by the plurality of insulating spacer structures **150** covering sidewalls of the plurality of bit line structures **140** and upper portions of second spaces between the plurality of insulating fences **180a**. The plurality of buried contacts **170** may fill lower portions of the first spaces and lower portions of the second spaces. Each of the plurality of buried contacts **170** may be connected to a corresponding one of the plurality of active regions **118**. Each of the plurality of landing pads **190** may extend to the upper portion of a corresponding one of the plurality of bit line structures **140**. The plurality of capacitor structures **200** may be formed to include the plurality of bottom electrodes **210**, the capacitor dielectric layer **220**, and the top electrode **230**. Each of the plurality of bottom electrodes **210** may be connected to a corresponding one of the plurality of landing pads **190**.

[0113] Since the plurality of insulating fences **180a** fill the plurality of fence holes **180Ha**, the first width **W1** that is the width of the lower portion of the plurality of insulating fences **180a** in the second horizontal direction (Y direction) between a pair of buried contacts **170** among the plurality of insulating fences **180a** may be less than the second width **W2** that is the width of the upper portion of the plurality of insulating fences **180a** in the second horizontal direction (Y direction) between a pair of landing pads **190**. The upper portions and the lower portions of the plurality of insulating fences **180a** may be portions more distant from the substrate **110** and portions closer to the substrate **110** in the vertical direction (Z direction) in a region where the plurality of buried contacts **170** overlap the plurality of landing pads **190**, respectively. In other words, in the plurality of insulating fences **180a**, the first width **W1** that is the width of the lower portion in the second horizontal direction (Y direction) may be less than the second width **W2** that is the width of the upper portion in the second horizontal direction (Y direction).

[0114] A width of the plurality of fence holes **180Ha** in the first horizontal direction (X direction) may be defined by a width between a pair of insulating spacer structures **150** facing each other, and may increase in the vertical direction (Z direction) perpendicular to the substrate **110**.

[0115] In the semiconductor memory device **3**, the second width **W2** of the upper portion of each of the plurality of insulating fences **180a** in the second horizontal direction (Y direction) may be greater than the first width **W1** of the lower portion of each of the plurality of insulating fences **180a** in the second horizontal direction (Y direction), and thus the plurality of buried contacts **170** between the lower portions of the plurality of insulating fences **180a** may have a relatively large horizontal width in the second horizontal direction (Y direction). Thus, the specific resistance of the plurality of buried contacts **170** may have a relatively small value, and the reliability of the electrical connection between the plurality of buried contacts **170** and the plurality of active regions **118** may increase.

[0116] FIGS. **20** and **21** are cross-sectional views sequentially illustrating a manufacturing method of the semiconductor memory device **3**, according to embodiments of the inventive concept. FIG. **20** is a cross-sectional view illustrating operations subsequent to an operation with reference to FIG. **8**, and descriptions given with reference to FIGS. **20** and **21** that are duplicates of descriptions given with reference to FIGS. **2** through **12** are omitted, and the same reference numerals may denote the same components.

[0117] Referring to FIGS. **8** and **20** together, by removing a portion of the mold layer **175** and a portion of the preliminary buried contact material layer **170p** by using the plurality of mask patterns **MK** as etching masks, a plurality of fence holes **180Hb** that causes the buried insulating layer **124** to be exposed to a bottom surface thereof may be formed. In some embodiment, the plurality of fence holes **180Hb** may be in a line in the first horizontal direction (X direction) and the second horizontal direction (Y direction), respectively. In an example embodiment, the mold layer **175** and the preliminary buried contact material layer **170p** may be patterned, by performing an etching process using the plurality of mask patterns **MK** as an etch mask, to form a plurality of

mold patterns separated from each other and the plurality of buried contacts **170** separated from each other, respectively.

[0118] The preliminary buried contact material layer **170p** may be separated into a plurality of buried contacts **170** by the plurality of fence holes **180Hb**.

[0119] In the process of forming the plurality of fence holes **180Hb**, due to differences in etching characteristics of the mold layer **175** and the preliminary buried contact material layer **170p**, a second width **W2a** that is a width of an upper portion of the plurality of fence holes **180Hb** between a pair of the mold layers **175** (i.e., a pair of the mold patterns) in the second horizontal direction (Y direction) may be less than a first width **W1a** that is a width of a lower portion of the plurality of fence holes **180Hb** between a pair of the buried contacts **170** in the second horizontal direction (Y direction). In other words, in the plurality of fence holes **180Hb**, the second width **W2a** that is the width of the upper portion in the second horizontal direction (Y direction) may be less than the first width **W1a** that is the width of the lower portion in the second horizontal direction (Y direction). In some embodiment, the first width **W1a** may be greater than the second width **W2a** by about 1 nm to about 5 nm.

[0120] A width of the plurality of fence holes **180Hb** in the first horizontal direction (X direction) may be defined by a width between a pair of insulating spacer structures **150** facing each other, and may increase in the vertical direction (Z direction) perpendicular to the substrate **110**.

[0121] In some embodiment, after the plurality of fence holes **180H** illustrated in FIG. **9** is formed, by further removing a portion of the buried contact **170**, the plurality of fence holes **180Hb** illustrated in FIG. **20** may be formed.

[0122] Referring to FIG. **21**, a plurality of insulating fences **180b** filling the plurality of fence holes **180Hb** may be formed. The plurality of insulating fences **180b** may include, for example, a nitride layer. After a fence material layer filling the plurality of fence holes **180Hb** and covering a top surface of the mold layer **175** is formed, the plurality of insulating fences **180b** may be formed by removing a portion of the fence material layer covering the top surface of the mold layer **175**.

[0123] The plurality of buried contacts **170** may be in spaces defined by the plurality of insulating spacer structures **150** that cover sidewalls of the plurality of bit line structures **140**, and by the plurality of insulating fences **180b**. For example, each of the plurality of buried contacts **170** may be disposed in a space defined by two adjacent insulating spacer structures facing each other across the space (e.g., cross-section A-A' of FIG. **21**) and by two adjacent insulating fences (e.g., cross-section C-C' of FIG. **21**). For example, the plurality of buried contacts **170** may be spaced apart from each other in the first horizontal direction (X direction) by the plurality of insulating spacer structures **150** and spaced apart from each other in the second horizontal direction (Y direction) by the plurality of insulating fences **180b**.

[0124] Each of the plurality of insulating fences **180b** may include an air gap **180AR** therein. In some embodiment, the air gap **180AR** may be inside the lower portion of each of the plurality of insulating fences **180b**. The lower portion of each of the plurality of insulating fences **180b** may be a portion thereof between a pair of buried contacts **170**.

[0125] FIG. **22** is a cross-sectional view of a semiconductor memory device **4**, according to example embodiments of the inventive concept. The semiconductor memory device **4** may correspond to the semiconductor device **10** of FIG. **1**.

[0126] Referring to FIG. **22**, the semiconductor memory device **4** may be formed by performing the processes described with reference to FIGS. **10** through **13**.

[0127] The semiconductor memory device **4** may include the substrate **110** having the plurality of active regions **118**. The plurality of gate dielectric layers **122**, the plurality of word lines **120**, and the plurality of buried insulating layers **124** may be sequentially formed inside the plurality of word line trenches **120T** formed in the substrate **110**. The element isolation layer **116** may define the plurality of active regions **118**. The plurality of insulating layer patterns (**112** and **114**) may be formed to cover the plurality of buried insulating layers **124**. The plurality of landing pads **190** may

be formed to fill the upper portions of first spaces defined by the plurality of insulating spacer structures **150** covering sidewalls of the plurality of bit line structures **140** and the upper portions of second spaces between the plurality of insulating fences **180b**. The plurality of buried contacts **170** may fill the lower portions of the first spaces and the lower portion of the second spaces. Each of the plurality of buried contacts **170** may be connected to a corresponding one of the plurality of active regions **118**. Each of the plurality of landing pads **190** may extend to the upper portion of a corresponding one of the plurality of bit line structures **140**. The plurality of capacitor structures **200** may be formed to include the plurality of bottom electrodes **210**, the capacitor dielectric layer **220**, and the top electrode **230**. The plurality of bottom electrodes **210** may be connected to the plurality of landing pads **190**.

[0128] Since the plurality of insulating fences **180b** fill the plurality of fence holes **180Hb**, the first width **W1a** that is a width of a lower portion of each of the plurality of insulating fences **180b** in the second horizontal direction (Y direction) between a pair of buried contacts **170** may be greater than the second width **W2a** that is a width of an upper portion of each of the plurality of insulating fences **180b** in the second horizontal direction (Y direction) between a pair of landing pads **190**. The upper portions and the lower portions of the plurality of insulating fences **180b** may be portions away from the substrate **110** and portions toward the substrate **110** in the vertical direction (Z direction) based on portions where the plurality of buried contacts **170** overlap the plurality of landing pads **190**, respectively. In other words, in the plurality of insulating fences **180b**, the first width **W1a** that is the width of the lower portion in the second horizontal direction (Y direction) may be greater than the second width **W2a** that is the width of the upper portion in the second horizontal direction (Y direction).

[0129] A width of the plurality of fence holes **180Hb** in the first horizontal direction (X direction) may be defined by a width between a pair of insulating spacer structures **150** facing each other, and may increase in the vertical direction (Z direction) perpendicular to the substrate **110**.

[0130] In the semiconductor memory device **4**, the second width **W2a** of the upper portion of each of the plurality of insulating fences **180b** in the second horizontal direction (Y direction) may be less than the first width **W1a** of the lower portion of each of the plurality of insulating fences **180b** in the second horizontal direction (Y direction), and thus, the plurality of landing pads **190** between the upper portions of each of the plurality of insulating fences **180a** may have a relatively large width in the second horizontal direction (Y direction). Thus, the specific resistance of the plurality of landing pads **190** may have a relatively small value, and the reliability of the electrical connection between the plurality of buried contacts **170** and the plurality of capacitor structures **200** may increase.

[0131] In addition, since the semiconductor memory device **4** includes the air gap **180AR** in each of the plurality of insulating fences **180b**, the parasitic capacitance of a space between a pair of buried contacts **170** having each of the plurality of insulating fences **180b** therebetween may decrease.

[0132] FIG. **23** is a cross-sectional view of a semiconductor memory device **5**, according to example embodiments of the inventive concept. The semiconductor device **5** may correspond to the semiconductor device **10** of FIG. **1**.

[0133] Referring to FIG. **23**, the semiconductor memory device **5** may include the substrate **110** having the plurality of active regions **118**. The plurality of gate dielectric layers **122**, the plurality of word lines **120**, and the plurality of buried insulating layers **124** may be sequentially formed inside the plurality of word line trenches **120T** formed in the substrate **110**. The element isolation layer **116** may define the plurality of active regions **118**. The plurality of insulating layer patterns (**112** and **114**) may be formed to cover the plurality of buried insulating layers **124**. The plurality of bit line structures **140a** may be formed on the plurality of insulating patterns (**112** and **114**). The plurality of landing pad **190** may be formed to the upper portions of first spaces defined by the plurality of insulating spacer structures **150** covering sidewalls of the plurality of bit line structures

140a and the upper portions of second spaces between the plurality of insulating fences **180a**. The plurality of buried contacts **170** may fill the lower portion of the first spaces and the lower portions of the second spaces. Each of the plurality of buried contacts **170** may be connected to a corresponding one of the plurality of active regions **118**. Each of the plurality of landing pads **190** may extend to the upper portion of a corresponding one of the plurality of bit line structures **140**. The plurality of capacitor structures **200** may be formed to include the plurality of bottom electrodes **210**, the capacitor dielectric layer **220**, and the top electrode **230**. Each of the plurality of bottom electrodes **210** may be connected to a corresponding one of the plurality of landing pads **190**. The plurality of landing pads **190** may cover at least a portion of a top surface of the plurality of expansion capping layers **158** of the bit line structure **140a**.

[0134] The semiconductor memory device **5** may include a plurality of insulating fences **180a** instead of the plurality of insulating fences **180** of the semiconductor memory device **2** illustrated in FIG. **16**. Accordingly, in the semiconductor memory device **5**, the second width **W2** of the upper portion of each of the plurality of insulating fences **180a** in the second horizontal direction (Y direction) may be greater than the first width **W1** of the lower portion of each of the plurality of insulating fences **180a** in the second horizontal direction (Y direction), and thus each of the plurality of buried contacts **170** between the lower portions of the plurality of insulating fences **180a** may have a relatively large width in the second horizontal direction (Y direction). Thus, the specific resistance of the plurality of buried contacts **170** may have a relatively small value, and the reliability of the electrical connection between the plurality of buried contacts **170** and the plurality of active regions **118** may increase.

[0135] FIG. **24** is a cross-sectional view of a semiconductor memory device **6**, according to example embodiments of the inventive concept. The semiconductor memory device **6** may correspond to the semiconductor device **10** of FIG. **1**.

[0136] Referring to FIG. **24**, the semiconductor memory device **6** may include the substrate **110** having the plurality of active regions **118**. The plurality of gate dielectric layers **122**, the plurality of word lines **120**, and the plurality of buried insulating layers **124** may be sequentially formed inside the plurality of word line trenches **120T** formed in the substrate **110**. The element isolation layer **116** may be formed to define the plurality of active regions **118**. The plurality of insulating layer patterns (**112** and **114**) may be formed to cover the plurality of buried insulating layers **124**. The plurality of bit line structures **140a** may be formed on the plurality of insulating patterns (**112** and **114**). The plurality of landing pads **190** may be formed to fill the upper portions of first spaces defined by the plurality of insulating spacer structures **150** covering sidewalls of the plurality of bit line structures **140a** and second spaces between the plurality of insulating spacer structures **150**. The plurality of buried contacts **170** may fill the lower portions of the first spaces and the lower portions of the second spaces. Each of the plurality of buried contacts **170** may be connected to a corresponding one of the plurality of active regions **118**. Each of the plurality of landing pads **190** may extend to the upper portion of a corresponding one of the plurality of bit line structures **140**. The plurality of capacitor structures **200** may be formed to include the plurality of bottom electrodes **210**, the capacitor dielectric layer **220**, and the top electrode **230**. Each of the plurality of bottom electrodes **210** may be connected to a corresponding one of the plurality of landing pads **190**. The plurality of landing pads **190** may cover at least a portion of a top surface of the plurality of expansion capping layers **158** of the bit line structure **140a**.

[0137] The semiconductor memory device **6** may include a plurality of insulating fences **180b** instead of the plurality of insulating fences **180a** of the semiconductor memory device **2** illustrated in FIG. **16**. Accordingly, in the semiconductor memory device **6**, since the second width **W2a** of the upper portion of each of the plurality of insulating fences **180b** in the second horizontal direction (Y direction) may be less than the first width **W1a** of the lower portion of each of the plurality of insulating fences **180b** in the second horizontal direction (Y direction), the plurality of landing pads **190** between the upper portions of each of the plurality of insulating fences **180a** may have a

relatively large width in the second horizontal direction (Y direction). Thus, the specific resistance of the plurality of landing pads **190** may have a relatively small value, and the reliability of the electrical connection between the plurality of buried contacts **170** and the plurality of capacitor structures **200** may increase.

[0138] In addition, since the semiconductor memory device **6** includes an air gap **180AR** in each of the plurality of insulating fences **180b**, the parasitic capacitance of a space (i.e., each of the plurality of insulating fences **180b**) between a pair of buried contacts **170** may be reduced.

[0139] While the present inventive concept has been particularly shown and described with reference to example embodiments thereof, the present inventive concept is not limited to the example embodiments, and various changes and modifications may be made by those of ordinary skill in the art without departing from the spirit and scope of the present inventive concept.

[0140] While the inventive concept has been particularly shown and described with reference to embodiments thereof, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the following claims.

Claims

1. A semiconductor memory device comprising: a substrate; a plurality of bit line structures extending in parallel with each other in a first horizontal direction on the substrate and spaced apart from each other in a second horizontal direction perpendicular to the first horizontal direction; a plurality of buried contacts between two adjacent bit line structures among the plurality of bit line structures and spaced apart from each other in the first horizontal direction; and a plurality of landing pads spaced apart from each other in the first horizontal direction on the plurality of buried contacts, each of the plurality of landing pads connected to a corresponding one of the plurality of buried contacts, wherein a first width of each of the plurality of buried contacts in the first horizontal direction is greater than a second width of a portion of each of the plurality of landing pads in the first horizontal direction between two adjacent bit line structures among the plurality of bit line structures.
2. The semiconductor memory device of claim 1, wherein a bottom surface of each of the plurality of landing pads is connected to a top surface of the corresponding one of the plurality of buried contacts to constitute a stepped surface.
3. The semiconductor memory device of claim 1, wherein the plurality of bit line structures include a plurality of bit lines and a plurality of insulating capping lines, wherein each of the plurality of insulating capping lines stacks on a corresponding one of the plurality of bit lines, and wherein a top surface of each of the plurality of buried contacts is at a level lower than an uppermost surface of each of the plurality of insulating capping lines.
4. The semiconductor memory device of claim 1, wherein each of the plurality of buried contacts includes polysilicon.
5. The semiconductor memory device of claim 1, wherein each of the plurality of landing pads includes metal, conductive metal nitride, or a combination of metal and conductive metal nitride.
6. The semiconductor memory device of claim 1, wherein the first width is greater than the second width by about 1 nm to about 5 nm.
7. The semiconductor memory device of claim 1, wherein each of the plurality of bit line structures includes a first metal-based conductive pattern, a second metal-based conductive pattern on the first metal-based conductive pattern, and an insulating capping line on the second metal-based conductive pattern.
8. The semiconductor memory device of claim 7, wherein the first metal-based conductive pattern includes titanium nitride (TiN) or TSN(Ti—Si—N), wherein the second metal-based conductive pattern includes W, or tungsten silicide (WSi.sub.x), and wherein the insulating capping line includes a silicon nitride layer.

9. A semiconductor memory device comprising: a substrate having a plurality of active regions defined by an element isolation layer; a plurality of bit line structures extending in parallel with each other in a first horizontal direction on the substrate and spaced apart from each other in a second horizontal direction perpendicular to the first horizontal direction, each of the plurality of bit line structures including a bit line and an insulating capping line stacked on the bit line; a plurality of first insulating spacer structures covering first sidewalls of the plurality of bit line structures; a plurality of second insulating spacer structures covering second sidewalls of the plurality of bit line structures; a plurality of insulating fences disposed in a first space between one of the plurality of first insulating spacer structures and one of the plurality of second insulating spacer structures adjacent thereto in the second horizontal direction and spaced apart from each other in the first horizontal direction; a plurality of buried contacts disposed in the first space and spaced apart from each other in the first horizontal direction, each of the plurality of buried contacts filling a lower portion of the first space and being connected to a corresponding one of the plurality of active regions, wherein each of the plurality of buried contacts and each of the plurality of insulating fences are alternately arranged in the first horizontal direction; and a plurality of landing pads spaced apart from each other in the first horizontal direction, each of the plurality of landing pads connected to a corresponding one of the plurality of buried contacts, each of the plurality of landing pads filling an upper portion of the first space and extending to an upper portion of a corresponding one of the plurality of bit line structures, wherein a first width of each of the plurality of buried contacts in the lower portion of the first space is greater than a second width of a portion of each of the plurality of landing pads in the upper portion of the first space, and the first width and the second width are measured in the first horizontal direction.

10. The semiconductor memory device of claim 9, wherein each of the plurality of insulating fences includes a nitride layer.

11. The semiconductor memory device of claim 9, wherein each of the plurality of first insulating spacer structures and the plurality of second insulating spacer structures includes a first insulating spacer, a second insulating spacer, and a third insulating spacer, and wherein each of the first insulating spacer and the third insulating spacer include a nitride layer, and the second insulating spacer includes an oxide layer.

12. The semiconductor memory device of claim 9, wherein each of the plurality of insulating fences includes a nitride layer.

13. The semiconductor memory device of claim 9, wherein a bottommost surface of each of the plurality of buried contacts is at a level lower than a top surface of each of the plurality of active regions.

14. The semiconductor memory device of claim 9, further comprising: a plurality of word lines filling lower portions inside a plurality of word line trenches formed in the substrate; and a plurality of buried insulating layers covering the plurality of word lines and filling upper portions inside the plurality of word line trenches, wherein the plurality of word line trenches cross the plurality of active regions and extend in parallel with each other in the second horizontal direction, and wherein a bottommost surface of each of the plurality of insulating fences is at a level lower than an uppermost surface of each of the plurality of buried insulating layers.

15. The semiconductor memory device of claim 9, wherein a bottom surface of each of the plurality of landing pads is connected to a top surface of the corresponding one of the plurality of buried contacts to constitute a stepped surface.

16. The semiconductor memory device of claim 9, wherein a top surface of each of the plurality of buried contacts is at a level lower than an uppermost surface of each of the plurality of insulating capping lines.

17. A semiconductor memory device comprising: a substrate having a plurality of active regions defined by an element isolation layer; a plurality of word line trenches crossing the plurality of active regions and extending in parallel with each other in a first horizontal direction; a plurality of

word lines filling lower portions inside the plurality of word line trenches; a plurality of buried insulating layers covering the plurality of word lines and filling upper portions inside the plurality of word line trenches; an insulating layer pattern covering the element isolation layer, the plurality of active regions, and the plurality of buried insulating layers; a plurality of bit line structures extending in parallel with each other in a second horizontal direction perpendicular to the first horizontal direction on the insulating layer pattern and spaced apart from each other in the first horizontal direction, each of the plurality of bit line structures including a bit line and an insulating capping line stacked on the bit line; a plurality of first insulating spacer structures covering first sidewalls of the plurality of bit line structures; a plurality of second insulating spacer structures covering second sidewalls of the plurality of bit line structures; a plurality of insulating fences disposed in a first space between one of the plurality of first insulating spacer structures and one of the plurality of second insulating spacer structures adjacent thereto in the first horizontal direction and spaced apart from each other in the second horizontal direction; a plurality of buried contacts disposed in the first space and spaced apart from each other in the second horizontal direction, each of the plurality of buried contacts filling a lower portion of the first space and being connected to a corresponding one of the plurality of active regions, wherein each of the plurality of buried contacts and each of the plurality of insulating fences are alternately arranged in the second horizontal direction; and a plurality of landing pads spaced apart from each other in the second horizontal direction, each of the plurality of landing pads connected to a corresponding one of the plurality of buried contacts, each of the plurality of landing pads filling an upper portion of the first space and extending to an upper portion of a corresponding one of the plurality of bit line structures, wherein a first width of each of the plurality of buried contacts in the lower portion of the first space is greater than a second width of a portion of each of the plurality of landing pads in the upper portion of the first space, wherein the first width and the second width are measured in the second horizontal direction, and wherein a bottom surface of each of the plurality of landing pads is connected to a top surface of the corresponding one of the plurality of buried contacts to constitute a stepped surface.

18. The semiconductor memory device of claim 17, wherein the insulating layer pattern includes a silicon oxide layer, a silicon nitride layer, a silicon oxynitride layer, or a combination thereof.

19. The semiconductor memory device of claim 17, wherein an uppermost surface of each of the plurality of buried contacts is at a level lower than an uppermost surface of each of the plurality of insulating capping lines, and wherein a bottommost surface of each of the plurality of buried contacts is at a level lower than a top surface of each of the plurality of active regions.

20. The semiconductor memory device of claim 17, wherein a bottommost surface of each of the plurality of insulating fences is at a level lower than an uppermost surface of each of the plurality of buried insulating layers.
